

SHODH — Criminal Network Analysis Platform (CNAP)

AI-Powered Law Enforcement Visual Intelligence Suite • 100% Air-Gapped Local Deployment • Zero Cloud & Zero External APIs

Operational Status: Production Ready | **Evidence Standard:** Indian Evidence Act §65B Admissible | **Security:** Completely Offline

1. Executive Summary

SHODH is an offline-capable visual intelligence platform engineered for state police departments, cybercrime cells, and anti-narcotics task forces to uncover, map, forecast, and dismantle multi-jurisdictional organized crime syndicates. It synthesizes heterogeneous investigative records—including Call Detail Records (CDRs), Hawala transactions, FIRs, field surveillance reports, and cell tower trails—into a coherent, interactive relationship graph.

All analytical computations, graph centrality algorithms, anomaly detectors, and time-series predictive forecasts run strictly on local hardware using **FastAPI**, **SQLite**, **NetworkX**, **ELK.js**, **D3.js**, and **React 19** without transmitting evidentiary records to any cloud vendor or third-party AI service.

2. Visual System Architecture

The architecture is organized into four distinct, air-gapped tiers ensuring courtroom evidentiary integrity, millisecond graph layouts, and strict surveillance warrant gating:

Tier 1: Presentation & Investigation	• Dual Graph Engines: ELK.js Hierarchical Link Graph (DAG) + Cytoscape Concentric Rings. • Analytical Views: D3 Past Chronology, D3 Predictive Timeline (12 Threat Forecasts). • Tactical GeoIntel: India-bounded Leaflet dark map with real-time radar sweep & meetups. • Interactive Modules: AI Investigator Assistant (RAG), Hierarchy Tree, Approvals Gate.	React 19 Vite ELK.js D3.js v7 Leaflet
Tier 2: Application & API Gateway	• Auth & RBAC: OAuth2 Password Flow + JWT (Investigator, Senior Authority, Admin). • REST API Engine: Multi-case endpoints, graph exports, pattern detectors, approvals. • Streaming Gateway: WebSocket telemetry manager for live suspect GPS pings.	FastAPI ASGI Uvicorn WebSockets Pydantic v2
Tier 3: Graph Intelligence & Local Analytics	• Multi-Factor Scoring: PageRank + Betweenness + Criminal Multiplier formula. • 12 Forensic Detectors: Hawala cycles, sudden bursts, burner SIM swapping, co-location. • Predictive Engine: Local time-series trend forecasting with legal uncertainty disclaimers. • Gated Traversal: Surveillance warrant gated BFS/DFS outer shell explorer.	NetworkX NumPy SciPy Evidence Builder
Tier 4: Evidentiary Storage & Persistence	• Relational Store: SQLite in WAL mode (Cases, Persons, CDRs, Hawala, FIRs, Waypoints). • In-Memory Topology: NetworkX MultiDiGraph for dynamic sub-graph extractions. • Audit Vault: Immutable evidentiary chain of custody logging every action.	SQLite 3 SQLAlchemy 2.0 Immutable Log

3. Visual Evidentiary Pipeline Flow

Investigation workflow from raw evidentiary ingestion to tactical surveillance and threat mitigation:

1. INGESTION	2. GRAPH SYNTHESIS	3. ANOMALY DETECTION	4. WARRANT DATE	5. TACTICAL ACTION
CDRs, Hawala Logs, FIRs, Tower Pings ingested via CSV/JSON (or Seed).	NetworkX constructs MultiDiGraph; computes PageRank & Suspicion Zones.	12 forensic detectors identify circular layering, bursts & burner hopping.	Senior Authority reviews & approves warrants for outer-shell expansion.	Live radar map tracking & 90-day predictive threat timeline intervention.

4. Law Enforcement 4-Colour Evidentiary Palette

Strictly calibrated for courtroom evidence readability and command center high-contrast standards:

■ White	#FFFFFF #F8F9FA	Dominant background canvas, neutral node fills, peripheral contacts.
■ Yellow	#FFD600 #FFF9C4	Primary investigation accent, Inner Concentric Core Zone (>75% Suspicion), key alerts.
■ Red	#E53935 #FFECEE	High-threat pattern alerts, criminal record badges, registered crime incident & FIR dots.
■ Black	#1A1A1A #333333	Evidentiary typography, high-contrast borders, timeline baseline, command framing.

5. Core Visual Intelligence Capabilities

Dual Network Graph Engines (Link Graph ELK & Concentric Rings)

- **Link Graph (ELK.js Layout Engine):** Hierarchical DAG layered flow with custom SVG CyberNodes and CyberEdges. Nodes dynamically display role badges, photo avatars, criminal history borders, and seed markers. Edges reveal ■ transaction sums and call counts dynamically on zoom. Features an interactive Minimap with viewport framing.
- **Concentric Suspicion Rings:** Court-compliant concentric zoning: **Inner Zone (>75%)** for syndicate kingpins; **Middle Zone (50–75%)** for coordinators; **Outer Zone (25–50%)** for peripheral couriers; and **Pruned (<25%)** innocent contacts rendered as discrete '?' placeholders.

Dual Evidentiary Timelines (Past Chronology vs Predictive Horizon)

- ■ **Past Timeline (D3.js):** Chronological timeline distinguishing white telecommunication pings from red registered FIR/crime incidents. Click any event for metadata and chain of custody.
- **Predictive Timeline:** 12 structured intelligence projections (Hawala settlement cycles, burner SIM rotations, courier handoffs) with interactive D3 time horizon slider (7 to 90 days), risk pill filters, and mandatory statutory evidentiary disclaimers.

AI Investigator Assistant

- Contextual natural language reasoning assistant querying local graph topology, suspect dossiers, alibis, and CDR logs without external API dependencies. Includes one-click quick investigative queries for hawala flows, bridge nodes, and score override audits.

Tactical GeoIntel Map

- India-bounded high-contrast dark map with SVG radar sweep animations, simulated live WebSocket GPS tracking, and physical meetup hotspots with concentric orange glow rings and attendee lists.

6. The 12 Forensic Pattern Detectors

1	Sudden Communication Burst	Call volume exceeding 3x the 30-day baseline prior to a crime incident.
2	New Unrelated Connections	Sudden direct calls between clusters with zero prior common associates.
3	Bridge Node (High Centrality)	Single entity connecting two or more mutually disjoint criminal groups.
4	Cross-Case Entity Linkage	Same phone, vehicle, or bank account appearing across multiple FIRs.
5	Unusual Location Sequence	Physically impossible transit velocities between distant cell towers.
6	Repeated Co-Location	Independent phones repeatedly sharing cell towers within 15-minute windows.
7	Structured Financial Smurfing	Multiple transactions placed just below mandatory reporting thresholds.
8	Circular Hawala Layering	Cyclic fund flows ($A \rightarrow B \rightarrow C \rightarrow A$) designed to obscure source of proceeds.
9	Call + Location Correlation	Short coordination calls immediately preceding co-located physical meetings.
10	Burner SIM / Identity Hopping	IMEI or SIM discarded sequentially while contacting the exact same core peers.
11	Sudden Community Formation	New sub-graph cluster emerging with rapid link density in a short window.
12	Dormant Network Reactivation	Inoperative criminal network showing sudden high-frequency reactivation.

7. Deployment & Default Credentials

One-Command Launch (Windows): Double-click run.bat or run cmd /c run.bat.

One-Command Launch (Linux / macOS): Run `chmod +x run.sh && ./run.sh`.

Backend API: <http://localhost:8000> • Frontend UI: <http://localhost:5173>

Investigator	investigator	invest123	Case analysis, graph traversal, score override with justification.
Senior Authority	senior	senior123	Surveillance warrant approvals, standing authorizations, override sign-off.
Administrator	admin	admin123	Immutable chain of custody audit logs, system administration.

8. Evidentiary Admissibility (Evidence Act §65B)

1. 100% Air-Gapped Operation: Zero telemetry or external cloud dependencies.

2. Immutable Evidentiary Audit Trail: Every node search, warrant request, and score override is permanently hashed and logged.

3. Deterministic Graph Math: Rankings are generated by transparent graph algorithms (PageRank, Betweenness) rather than non-reproducible black-box models.

4. Surveillance Warrant Gating: Peripheral contacts cannot be probed without senior officer sign-off.